

Zainab Sharif

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Summary

Software Engineering student at FAST-NUCES. Demonstration of strong object-oriented design and software engineering principles through independently developed graphical applications. Combines technical development with a background in digital graphic design, team collaboration, and leadership through extracurricular roles.

Technical Skills

Programming Languages: C++, C, Python, SQL (MSSQL), Assembly Language, HTML, CSS

Software Engineering: Object-Oriented Programming, SOLID Principles, Software Design Patterns, UML, Data Structures, File I/O

Libraries & Frameworks: SFML (Graphics, Audio, Window), Flask, SQL Database Integration

Design & Collaboration: Canva (Game Assets, UI Mockups, Social Media Graphics), Git, Visual Studio

Soft Skills: Team Collaboration, Project Documentation, Graphic Design, Leadership

Education

Bachelor of Science in Software Engineering, FAST-NUCES, Lahore

- **Relevant Coursework:** Data Structures, Object Oriented Programming, Database Systems, Operating Systems, Computer Organization and Assembly Language, Software Requirements Engineering, Software Design and Architecture

Intermediate from Punjab College, Lahore

Projects

Tic Tac Toe – Arcade Style — C++, SFML

- Built a fully graphical Tic Tac Toe game using C++ and SFML with a custom arcade visual style
- Designed all background graphics and UI assets from scratch using Canva for a polished, original look
- Implemented an interactive player selection screen allowing users to choose between playing as X or O
- Developed responsive mouse-driven input handling for seamless board interaction using SFML's event system
- Implemented complete game-state logic to detect and display all three outcomes: win, loss, and draw
- Integrated background music to enhance the arcade atmosphere using the SFML audio module

Nexus E-Library Management System — Python, Flask, MSSQL

- Built a full-stack research paper management platform with a Flask backend and Single-Page Application frontend connected to Microsoft SQL Server
- Implemented JWT-based authentication with bcrypt password hashing and role-based access control across Admin, Researcher, and Student roles
- Developed a plagiarism detection engine using N-gram analysis and Jaccard similarity, automatically flagging uploads that exceed a configurable similarity threshold
- Engineered a personalized recommendation system that analyzes user activity and bookmarks to suggest relevant papers, with a cold-start fallback for new users
- Designed RESTful API endpoints covering paper CRUD, multi-parameter search, bookmarking, activity tracking,

and admin user management

FLEX-Like University Management System

- Building a university portal with all core FLEX features including student registration, course enrollment, grade management, and timetable scheduling
- Adding an automated attendance sub-system to eliminate roll-call based attendance with a more efficient and scalable alternative

Retro Style Pixel Hangman — C++ , SFML

- Developed a fully graphical Hangman game from scratch using C++ and SFML with a retro pixel arcade aesthetic
- Designed and produced all custom background art and visual assets independently using Canva
- Implemented a multi-category word selection system, allowing players to choose from distinct thematic categories at game start
- Engineered a file handling system to load word lists from external data files, enabling category expansion without recompilation
- Implemented real-time tracking and on-screen display of incorrect letter guesses, remaining attempts, and partial word progress
- Rendered dynamic win and lose condition screens with contextual messaging upon game completion

Social Media Platform — C++ , MSSQL

OOP Project

- Designed and implemented a console-based social media simulation applying core object-oriented programming principles
- Applied encapsulation to protect and manage user and post data through well-defined class interfaces
- Leveraged inheritance to model distinct user types and post categories with shared and specialized behaviors
- Integrated an MSSQL database backend for persistent storage and retrieval of user and post records
- Built a menu-driven interface for intuitive navigation of features including posting, following, and feed viewing

Chess with GUI — C++ , SFML

- Developed a fully graphical Chess game with functional board and piece movement system using C++ and SFML
- Implemented core chess logic including move validation, turn management, and coordinate-based board mapping
- Designed an interactive GUI with mouse-driven piece selection and placement using SFML's event handling system
- Engineered game state management using arrays and conditional logic to track and update board positions across turns

Extracurricular Activities

Graphic Design Officer – Career Counselling Society, FAST-NUCES

- Designed promotional content including custom reel covers, Instagram posts, and event banners for society outreach campaigns
- Maintained consistent visual branding across all digital platforms and coordinated with organizing teams on event material

Graphic Design Officer – ACM Student Chapter, FAST-NUCES

- Produced visual content for ACM chapter events and technical competitions including social media graphics and promotional banners

- Collaborated with organizing teams to deliver design assets aligned with event themes and deadlines